

4. VISUALIZATIONS

We propose 8 visualizations directly aligned with the client's requirements — 5 committed deliverables and 3 stretch goals. Each targets specific stakeholder needs and analysis goals. Numbering follows earlier visualization IDs; committed vs. stretch priority is defined by grouping, not numeric order.

Committed Visualizations

4.1 Interactive Organ Dashboard (Landing Page)

Type: Multi-panel dashboard with organ selector

Stakeholders: All groups

Selecting an organ shows KPI cards (publications, experts, datasets, funding projects) with sparklines. Serves as the entry point to all other views.

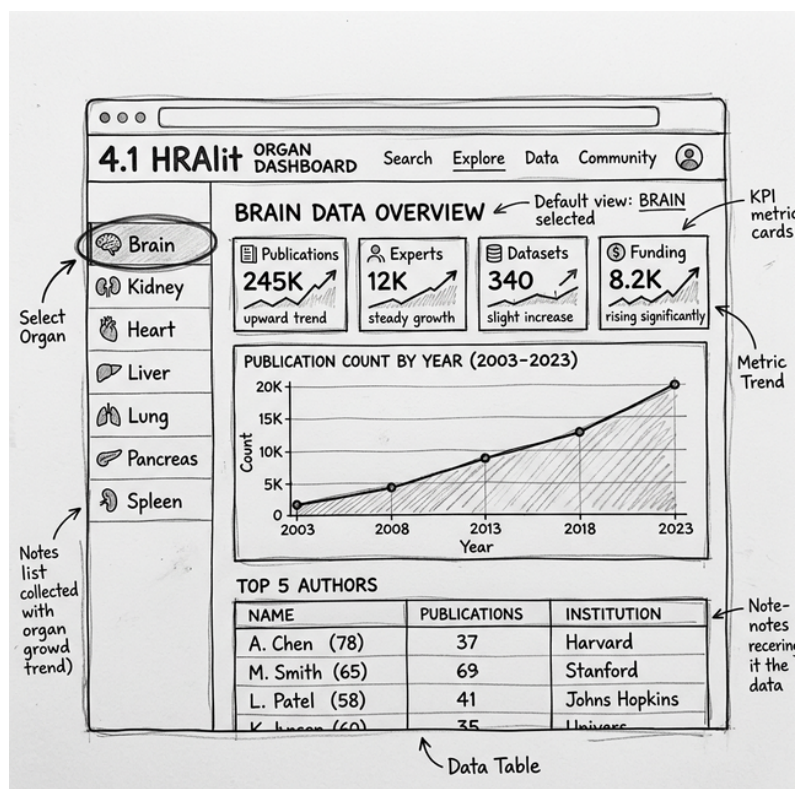


Figure 2. Sketch of Interactive Organ Dashboard (Viz 4.1).

4.2 Publication & Funding Trend Lines with 5-Year Prediction

Type: Line chart with forecasting band

Stakeholders: Funding Agencies, Policy Makers, HRA Team

Past 20 years of publication and funding counts per organ as solid lines, plus dashed forecast lines for 2024–2028 with shaded 95% confidence intervals. Users can toggle between organs.

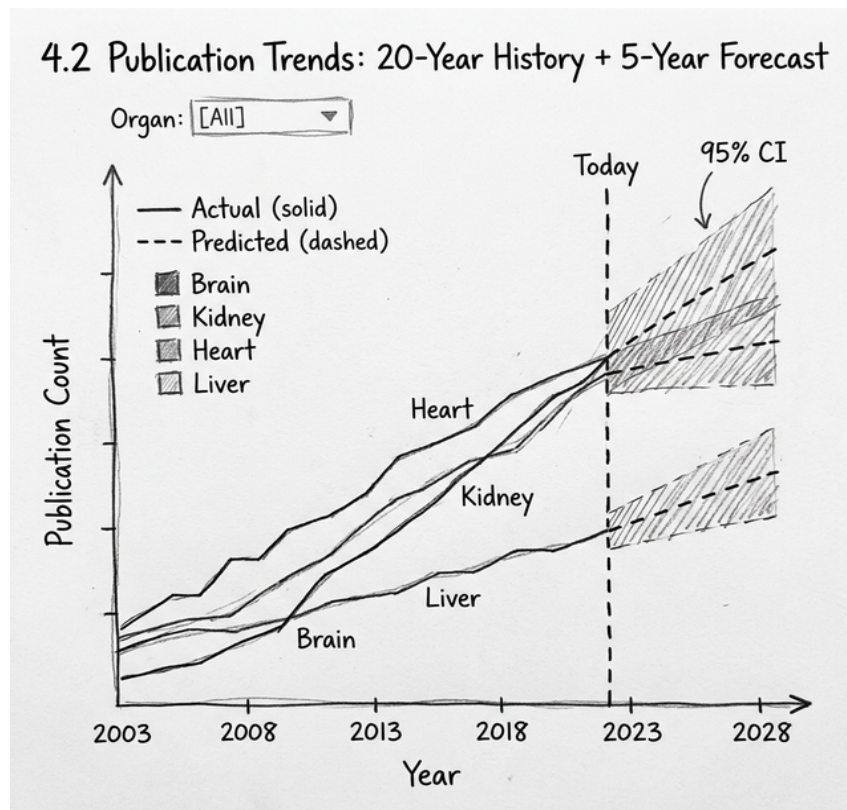


Figure 3. Sketch of Publication & Funding Trend Lines with Forecast (Viz 4.2).

4.3 USA–China Collaboration Over Time

Type: Grouped bar chart + trend line

Stakeholders: Policy Makers, Funding Agencies

Year-by-year count of co-authored publications between US and Chinese institutions, with comparison lines for US–UK and US–Germany. Shows how international cooperation has evolved.

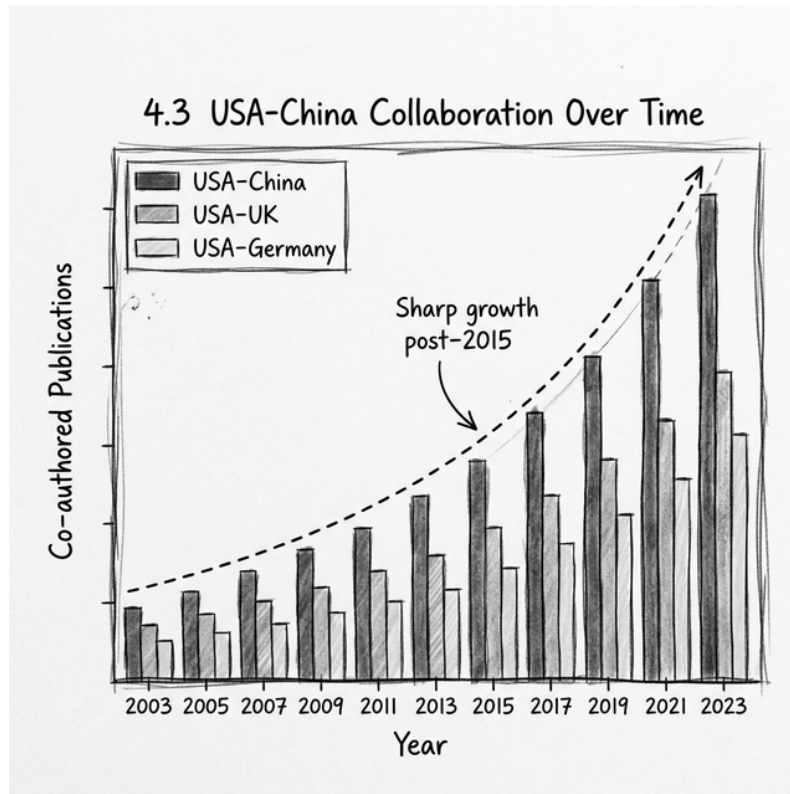


Figure 4. Sketch of USA–China Collaboration Over Time (Viz 4.3).

4.4 Temporal Co-authorship Network (Key Nodes)

Type: Force-directed network with time slider

Stakeholders: Researchers, HRA Team

Interactive network graph showing co-authorship clusters for a selected organ. A time slider divides data into 5-year windows (2003–2007, 2008–2012, 2013–2017, 2018–2023). Node size = publication count; color = institution/country. Networks limited to top 200–500 authors per organ for readability.

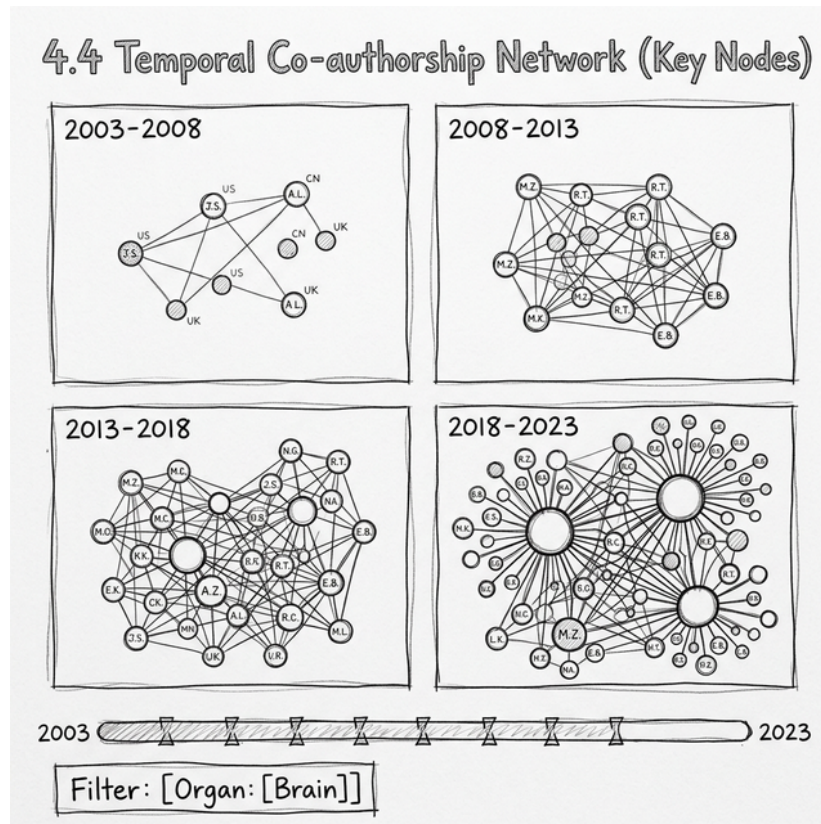


Figure 5. Sketch of Temporal Co-authorship Network (Viz 4.4).

4.6 Top Funders by Country Cooperation

Type: Grouped horizontal bar chart / heatmap

Stakeholders: Funding Agencies, Policy Makers

For each major country (USA, China, UK, Germany, Japan), shows the top 5 funders whose grants most frequently appear in internationally co-authored publications.

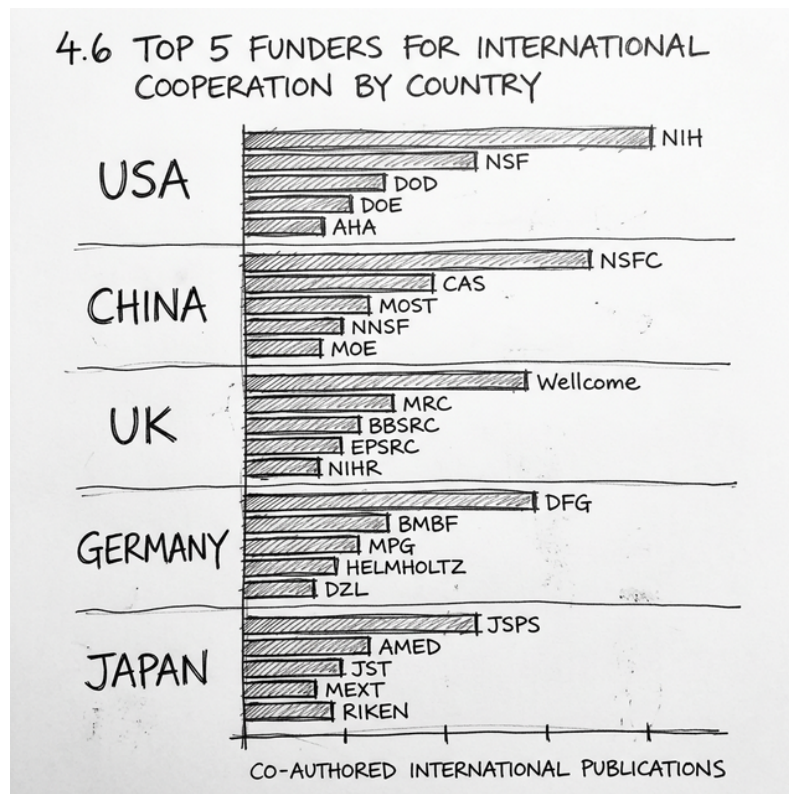


Figure 6. Sketch of Top Funders by Country Cooperation (Viz 4.6).

Stretch Goals

4.5 Grant-Linkage Sankey — Funder → Organ → Outputs

Type: Sankey / alluvial diagram

Stakeholders: Funding Agencies, Policy Makers

Traces grant-linkage frequency from funder → organ → research outcomes. Band width represents the number of co-linked publications, not dollar amounts.

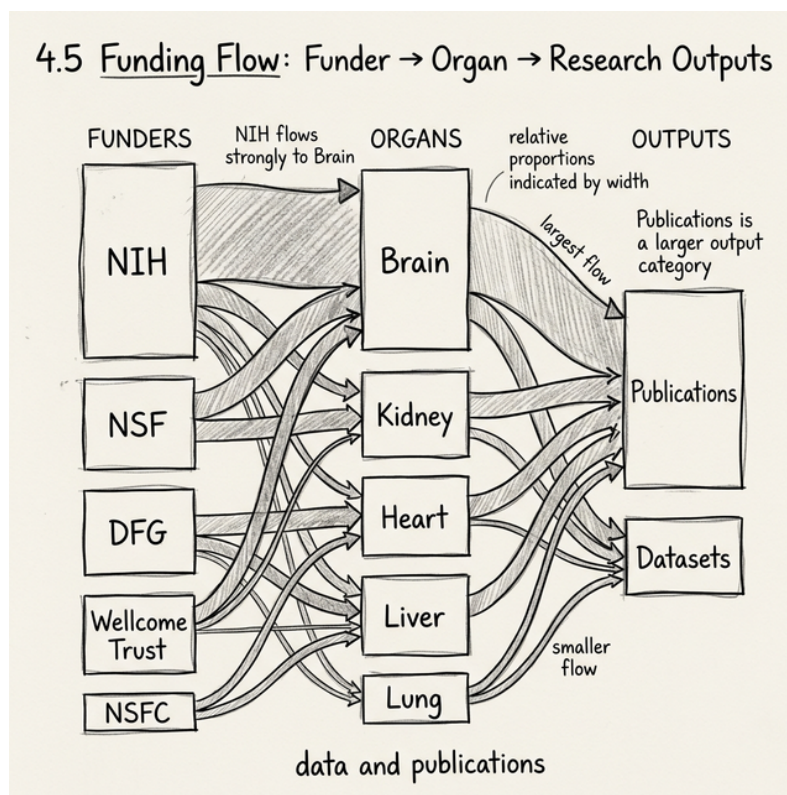


Figure 7. Sketch of Grant-Linkage Sankey (Viz 4.5).

4.7 Geo Map — Global Expertise & Funding Distribution

Type: Choropleth + proportional symbol map

Stakeholders: Policy Makers, Researchers

World map showing institution locations (sized by author count) and funder origins. Filterable by organ. Highlights geographic concentrations and gaps, with special emphasis on the USA–China research axis.

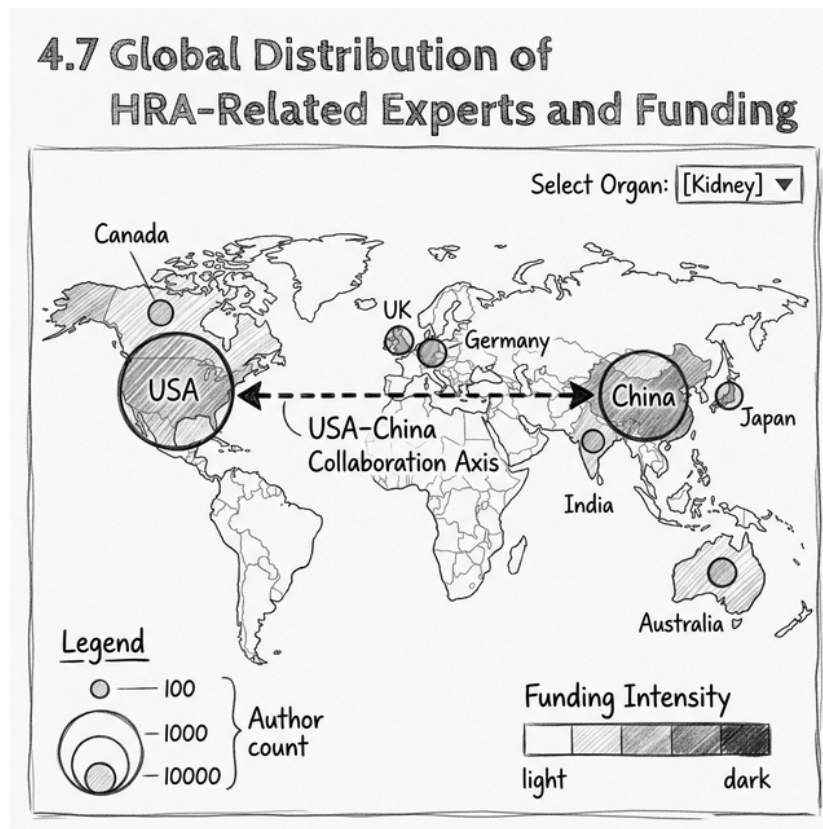


Figure 8. Sketch of Global Expertise & Funding Map (Viz 4.7).

4.8 University–Funder Relationship Heatmap

Type: Matrix heatmap

Stakeholders: Funding Agencies, Researchers

Rows = top institutions, columns = top funders. Cell intensity = number of co-linked publications. Reveals which university–funder pairings are strongest.

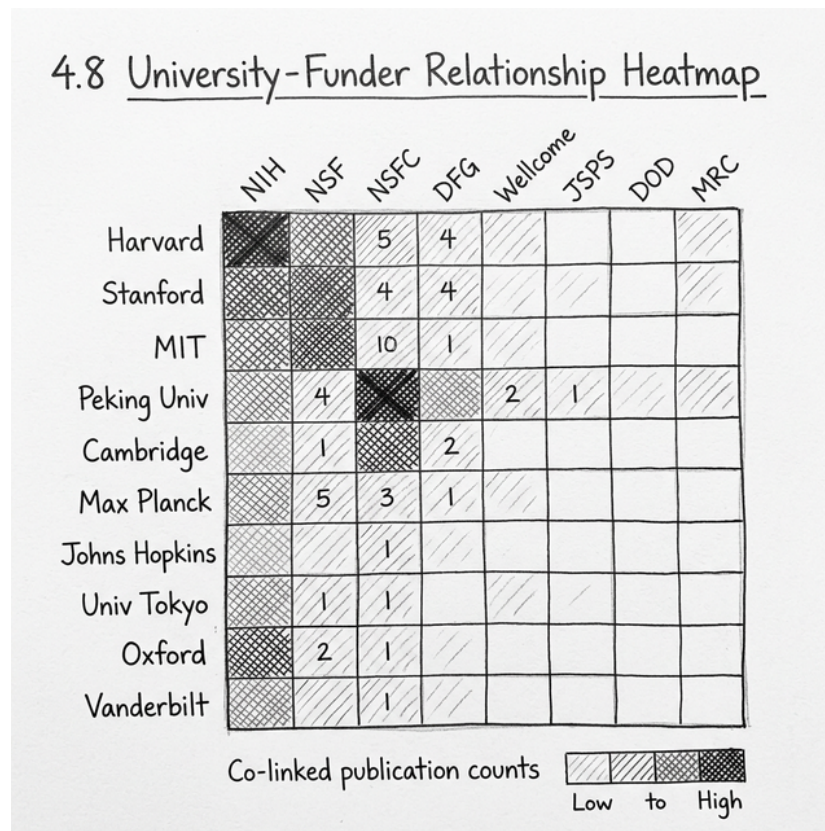


Figure 9. Sketch of University–Funder Relationship Heatmap (Viz 4.8).